

A RESIDUAL-CLASS CRITERION FOR EXCLUDING TYPE I NAVIER–STOKES SINGULARITIES

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ABSTRACT. We give a conditional reduction for Type I singularities of finite-energy suitable weak solutions of the three-dimensional Navier–Stokes equations. Starting from the positive pressure-normalized Caffarelli–Kohn–Nirenberg concentration carried by every singular point, we extract ancient limits by the Seregin blow-up procedure and separate the possible centered limits into small-amplitude, stationary L^3 , uniformly L^3 -tight, fast-decaying, structured, and residual alternatives. The small-amplitude and stationary L^3 alternatives are excluded unconditionally by classical Liouville theorems. The structured taxonomy also includes the established Lei–Zhang–Zhao $L_t^\infty L_x^p$ -swirl branch and the Lei–Ren–Zhang periodic bounded-swirl branch. Assuming Liouville criteria for the remaining tight and structured alternatives, full finite-energy Type I exclusion is reduced to one explicit analytic assertion: the residual non-tight/non-decaying class is empty, or else satisfies a replacement Liouville theorem. The positive concentration and the local Type I/Type II dichotomy are invoked as results proved earlier in the series; the additional inputs are precisely the stated nonclassical Liouville criteria.

1. INTRODUCTION

Let (u, p) be a finite-energy suitable weak solution of the three-dimensional incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$,

$$(1.1) \quad \partial_t u + u \cdot \nabla u + \nabla p = \Delta u, \quad \nabla \cdot u = 0.$$

A finite-time singular point $z_* = (x_*, T)$ is called Type I, in the sense used here, if the scale-invariant L^∞ quantity remains bounded near T :

$$(1.2) \quad \limsup_{t \uparrow T} \sqrt{T-t} \|u(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Equivalently, after decreasing the time interval if necessary, there are constants $M < \infty$ and $t_0 < T$ such that

$$\|u(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} \leq M(T-t)^{-1/2} \quad \text{for } t_0 < t < T.$$

This is the form used when passing the Type I bound to blow-up sequences. Indeed, if the limsup in (1.2) equals $L < \infty$, choose $M > L$; by the definition of limsup there is $t_0 < T$ such that $\sqrt{T-t} \|u(t)\|_{L^\infty} \leq M$ for $t_0 < t < T$. The local Caffarelli–Kohn–Nirenberg concentration theorem gives a sequence of shrinking backward parabolic cylinders centered at every singular point on which the pressure-normalized critical density is bounded from below. In the Type I case, the Seregin blow-up procedure applied to such a concentration sequence produces a nonzero bounded ancient solution in backward self-similar variables,

$$(1.3) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \nabla \cdot V = 0,$$

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defined on $\mathbb{R}^3 \times \mathbb{R}$. One first extracts an unrenormalized ancient suitable weak limit and then passes to centered variables,

$$V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t), \quad \tau = -\log(-t).$$

The inherited positive local critical density prevents the centered limit from vanishing identically.

Remark 1.1 (Source-level critical L^3 regime is already closed). If the original finite-energy suitable weak solution satisfies

$$\sup_{0 < t < T} \|u(t)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then the endpoint theorem of Escauriaza–Seregin–Sverak [3] implies that no point of $\mathbb{R}^3 \times \{T\}$ is singular. Thus a Type I singularity would have to lie outside the global critical $L_t^\infty L_x^3$ source regime. This is a condition on the original solution, not on an individual ancient limit, and it is therefore not included among the residual classes below.

No unconditional Liouville theorem is known for all bounded ancient solutions of (1.3). The reduction below is therefore organized around classes for which Liouville mechanisms are available or are formulated below as explicit hypotheses. Very small bounded ancient solutions vanish by the mild formulation and decay of the centered Stokes semigroup. Stationary self-similar profiles in $L^3(\mathbb{R}^3)$ vanish by the theorem of Necas–Ruzicka–Sverak. Uniform L^3 -tightness is the natural compactness condition for minimal-element and compact-hull arguments. Fast decay and geometric structure give additional routes to rigidity. What remains is the residual non-tight/non-decaying class. The main theorem proves that excluding this residual class, together with the stated Liouville inputs for the nonclassical classes, excludes finite-energy Type I singularities.

1.1. Logical ledger of the reduction. The argument below is a finite exhaustion argument. It relies on the following inputs. First, local CKN concentration at a singular point supplies a positive pressure-normalized density. Second, the Type I bound and Seregin compactness yield a nonzero bounded ancient solution of (1.3). Third, the source-level $L_t^\infty L_x^3$ regime has already been removed by Escauriaza–Seregin–Sverak. Fourth, small ancient limits and stationary L^3 ancient limits are unconditionally zero. Fifth, the tight, structured, and residual classes are eliminated precisely under the hypotheses explicitly stated below. Thus the final theorem is conditional precisely in those places where a nonclassical Liouville theorem is still required.

2. LOCAL CONCENTRATION AND SEREGIN ANCIENT LIMITS

Definition 2.1 (Finite-energy Type I source). A pair (u, p) is a finite-energy Type I source at $z_* = (x_*, T)$ if (u, p) is a suitable weak solution of (1.1) on $\mathbb{R}^3 \times (0, T)$ in the Caffarelli–Kohn–Nirenberg sense,

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)), \quad p \in L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (0, T)),$$

the local energy inequality holds, z_* is a singular point, and the Type I bound (1.2) holds. When the singular point is part of the notation, we write (u, p, z_*) for the corresponding source triple.

Here a point is regular if the velocity is essentially bounded in some backward parabolic cylinder with that point as top center; otherwise it is singular. This is the regularity notion used in the CKN epsilon-regularity criterion.

For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

For an integrable function f on a ball B , write

$$(f)_B = \frac{1}{|B|} \int_B f(x) dx.$$

Define the pressure-normalized CKN quantities

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt$$

and

$$D(z_0, r) = r^{-2} \int_{t_0-r^2}^{t_0} \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

Throughout the paper, pressures are understood modulo functions of time. Thus two pressures p and $p - b(t)$ represent the same velocity equation and the same local energy inequality. All pressure norms used below are either pressure oscillations over balls or are written after subtracting an explicitly allowed time-dependent gauge. This convention is the usual CKN pressure normalization. Indeed, subtracting a function of time has zero spatial gradient, leaves the distributional equation and the local energy inequality unchanged, and cancels from every pressure oscillation $p - (p)_{B_r}(t)$.

Theorem 2.2 (Local concentration at singular points). *Let (u, p) be suitable in a parabolic neighborhood of $z_* = (x_*, T)$. If z_* is singular, then there are $\eta > 0$ and radii $r_k \downarrow 0$, with $Q_{r_k}(z_*)$ contained in the neighborhood, such that*

$$(2.1) \quad C((x_*, T), r_k) + D((x_*, T), r_k) \geq \eta \quad \text{for all } k.$$

Moreover, along every such positive local concentration sequence, either a local Type I scale-invariant L^∞ bound holds near (x_*, T) , or the sequence lies in the local Type II alternative.

Proof. Let $\varepsilon_{\text{CKN}} > 0$ be the absolute epsilon-regularity threshold for the pressure-normalized CKN criterion [2, 6]. If $C(z_*, r) + D(z_*, r) < \varepsilon_{\text{CKN}}$ along a sequence $r \downarrow 0$, then u is locally bounded in a smaller cylinder and z_* is regular. Since z_* is singular, there are $\eta > 0$ and radii $r_k \downarrow 0$ for which (2.1) holds. Along these same radii, either the scale-invariant quantity

$$\sup_{Q_{\theta r_k}(z_*)} (T - t)^{1/2} |u|$$

is bounded at some fixed local scale $\theta \in (0, 1)$, which is the local Type I alternative, or no such uniform bound is available along the shrinking cylinders. The latter case is precisely the local Type II alternative. This uses only the CKN criterion and the definitions of the two local rate alternatives. $\square$

Lemma 2.3 (Scaling and pressure gauges). *Let (u, p) be suitable near $z_* = (x_*, T)$, and set*

$$u^{(\lambda)}(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \quad p^{(\lambda)}(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

Then $(u^{(\lambda)}, p^{(\lambda)})$ is suitable on the rescaled domain. Moreover

$$C_{u^{(\lambda)}}((0, 0), r/\lambda) = C_u(z_*, r), \quad D_{p^{(\lambda)}}((0, 0), r/\lambda) = D_p(z_*, r),$$

where the pressure averages are taken over the corresponding balls. Replacing $p^{(\lambda)}$ by $p^{(\lambda)} - b(t)$, with b depending only on time, does not change the equation, the local energy inequality, or the quantity D .

Proof. The distributional equation and the local energy inequality are invariant under the Navier–Stokes parabolic scaling after testing with pulled-back nonnegative cutoffs. The cylinder Jacobian is λ^5 , while $|u|^3$, $|p - (p)_{B_{\lambda r}}|^{3/2}$, and the prefactor r^{-2} scale as λ^3 , λ^3 , and λ^{-2} , respectively. Consequently C and D are unchanged by the transformation. If $p^{(\lambda)}$ is replaced by $p^{(\lambda)} - b(t)$, then the additional term has zero spatial gradient and its contribution to the local energy identity vanishes because $u^{(\lambda)}$ is divergence free. It also cancels from the oscillation defining D . $\square$

Proposition 2.4 (Extraction of a normalized Seregin limit). *Let (u, p, z_*) be a finite-energy Type I source. Then some positive concentration sequence $\lambda_k \downarrow 0$ produces, after passing to a subsequence and fixing pressure gauges, an ancient suitable weak limit (U, Π) on $\mathbb{R}^3 \times (-\infty, 0)$. The limit satisfies*

$$\sup_{t < 0} \sqrt{-t} \|U(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Its centered representative $V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t)$, $\tau = -\log(-t)$, is a Seregin ancient limit satisfying (2.2).

Proof. Choose $\lambda_k \downarrow 0$ from Theorem 2.2. The rescaled domains exhaust $\mathbb{R}^3 \times (-\infty, 0)$, and Theorem 2.3 preserves suitability, the CKN normalization, and pressure gauges. The extraction itself is the classical Seregin blow-up construction [10], in the local Type I formulation of Albritton–Barker [1].

The estimates used by that construction in the present normalization are the following standard local compactness estimates. The Type I bound gives uniform compact-cylinder L^∞ control of the rescaled velocities. The local energy inequality gives uniform local $L_t^\infty L_x^2$ and $L_t^2 \dot{H}_x^1$ bounds. After subtracting spatial pressure averages on fixed balls, the pressure is bounded in $L_{\text{loc}}^{3/2}$. The equation then bounds $\partial_t u^{(k)}$ in a negative Sobolev space, so the Aubin–Lions–Simon lemma gives strong local velocity convergence, while the pressure converges weakly in $L_{\text{loc}}^{3/2}$. These convergences pass the equation, divergence-free constraint, and local energy inequality to an ancient suitable weak limit; standard local regularity under the inherited Type I bound gives smoothness away from the terminal time.

The point specific to the present formulation is the normalization used below. The positive local CKN density is supplied by Theorem 2.2. Its persistence follows from choosing the concentration sequence so that a fixed subcylinder carries positive pressure-normalized density and then using the strong local L^3 convergence of velocities together with weak lower semicontinuity for the gauged $L^{3/2}$ -pressure norm. Under the change of variables $(x, t) = (e^{-\tau/2}y, -e^{-\tau})$, the compact negative-time cylinder carrying positive mass maps to $K \Subset \mathbb{R}^3 \times \mathbb{R}$, and the pressure gauge becomes a function $a(\tau)$. This is precisely (2.2). $\square$

Definition 2.5 (Seregin ancient limit). A smooth bounded ancient solution V of (1.3), with pressure understood modulo functions of τ , is called a Seregin ancient limit if there exist a finite-energy Type I source (u, p, z_*) , scales $\lambda_k \downarrow 0$ forming a positive local concentration sequence in the sense of (2.1), and the corresponding rescalings

$$u^{(k)}(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t), \quad p^{(k)}(x, t) = \lambda_k^2 p(x_* + \lambda_k x, T + \lambda_k^2 t),$$

which converge locally, after passing to a subsequence and choosing pressure gauges, to an ancient suitable weak solution (U, Π) on $\mathbb{R}^3 \times (-\infty, 0)$. The compactness is taken along a concentration subsequence for which the positive pressure-normalized CKN density persists in the limit. The limit U satisfies

$$\sup_{t < 0} \sqrt{-t} \|U(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} < \infty,$$

and V is the self-similar representative of U :

$$V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t), \quad \tau = -\log(-t).$$

The associated centered pressure class is represented by

$$P(y, \tau) = (-t) \Pi(y\sqrt{-t}, t),$$

and replacing Π by $\Pi - b(t)$ replaces P by $P - e^{-\tau} b(-e^{-\tau})$. Thus the pressure remains defined modulo a function of τ . In particular, after passing to centered variables, there are a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, a pressure gauge $a(\tau)$, and $\eta_0 > 0$ such that

$$(2.2) \quad \int_K \left(|V(y, \tau)|^3 + |P(y, \tau) - a(\tau)|^{3/2} \right) dy d\tau \geq \eta_0.$$

Definition 2.6 (Normalized Seregin collection). A collection $\mathcal{S}$ of Seregin ancient limits is called normalized if there are a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and a constant $\eta_0 > 0$ such that every $V \in \mathcal{S}$, with its associated centered pressure P , admits a measurable time-dependent pressure gauge $a_V(\tau)$ for which

$$(2.3) \quad \int_K \left(|V(y, \tau)|^3 + |P(y, \tau) - a_V(\tau)|^{3/2} \right) dy d\tau \geq \eta_0.$$

When $\mathcal{S}$ is obtained from one finite-energy Type I source, K and η_0 are fixed by the chosen concentration normalization; the admissible gauge is allowed to depend on the element V . The condition is independent of the representative of the pressure class: replacing P by $P - c(\tau)$ only replaces the gauge $a_V(\tau)$ by $a_V(\tau) - c(\tau)$. The integral is finite on K for every Seregin limit because local $L^{3/2}$ -pressure compactness after admissible gauges is part of the extraction, and the centered change of variables has bounded Jacobian and bounded pressure scaling factor on every compact τ -interval.

Definition 2.7 (Associated normalized collection). Fix a finite-energy Type I source (u, p, z_*) . An associated normalized collection is obtained as follows. First choose one positive local concentration normalization, namely a compact cylinder and a lower bound which persist in the ancient limit by the scale selection and compactness argument in [Theorem 2.4](#). Then let $\mathcal{S}$ be any collection of Seregin ancient limits produced from the same source by subsequences compatible with that fixed normalization, each equipped with an admissible centered pressure gauge. The common cylinder K and lower bound η_0 are those fixed in the first step. With these data, $\mathcal{S}$ satisfies [Theorem 2.6](#). The gauges are measurable: before centering they are spatial averages over fixed balls, and after centering they are composed with the smooth time map $t = -e^{-\tau}$ and multiplied by the smooth factor $e^{-\tau}$.

Remark 2.8. The nonvanishing conditions (2.2) and (2.3) are not additional assumptions. They are the persistence, under the compactness used in the Seregin blow-up procedure, of the positive CKN concentration supplied by [Theorem 2.2](#). This persistence is precisely the scale-selection and nonzero-limit mechanism in [Theorem 2.4](#). The sole choice made in forming $\mathcal{S}$ is which concentration normalization is fixed once and for all.

Lemma 2.9 (Centered equation). *Let (U, Π) be a smooth ancient Navier–Stokes solution on $\mathbb{R}^3 \times (-\infty, 0)$, and define*

$$t = -e^{-\tau}, \quad x = e^{-\tau/2} y, \quad V(y, \tau) = e^{-\tau/2} U(e^{-\tau/2} y, -e^{-\tau}),$$

with

$$P(y, \tau) = e^{-\tau} \Pi(e^{-\tau/2} y, -e^{-\tau}).$$

Then V satisfies (1.3) and $\nabla_y \cdot V = 0$.

Proof. Differentiate $U(x, t) = e^{\tau/2} V(e^{\tau/2} x, \tau)$, where $\tau = -\log(-t)$. The terms $\partial_t U$, $(U \cdot \nabla)U$, $\nabla \Pi$, and ΔU all carry the common factor $e^{3\tau/2}$ after expressing them in (y, τ) . The derivative of the self-similar variables contributes $-\frac{1}{2}(V + y \cdot \nabla V)$, and the pressure transformation gives $\nabla_y P$. Since $\nabla_x \cdot U = e^\tau \nabla_y \cdot V$, the divergence-free condition is also preserved. Dividing the transformed equation by the common factor gives (1.3). $\square$

Lemma 2.10 (Zero velocity and pressure normalization). *Let $V \equiv 0$ solve (1.3) in distributions on $\mathbb{R}^3 \times \mathbb{R}$. Then every associated pressure satisfies $\nabla_y P = 0$. Consequently, after subtracting the admissible gauge given by the resulting spatial constant $a(\tau)$, the integrand in (2.2), and likewise the integrand in (2.3), vanishes identically.*

Proof. With $V \equiv 0$, equation (1.3) reduces to $\nabla_y P = 0$ in distributions, meaning

$$\int_{\mathbb{R}} \int_{\mathbb{R}^3} P(y, \tau) \nabla_y \cdot \varphi(y, \tau) dy d\tau = 0$$

for every $\varphi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R}; \mathbb{R}^3)$. By applying this identity to test fields of the form $\varphi(y, \tau) = \psi(\tau)\phi(y)e_j$, with $\psi \in C_c^\infty(\mathbb{R})$ and $\phi \in C_c^\infty(\mathbb{R}^3)$, one obtains

$$\int_{\mathbb{R}} \psi(\tau) \langle \partial_{y_j} P(\cdot, \tau), \phi \rangle d\tau = 0.$$

Since this holds for every ψ , Fubini's theorem and the fundamental lemma of distributions imply that $\partial_{y_j} P(\cdot, \tau) = 0$ in $\mathcal{D}'(\mathbb{R}^3)$ for almost every τ . Intersecting the three full-measure sets gives $\nabla_y P(\cdot, \tau) = 0$ for almost every τ . The elementary distributional fact that a distribution on $\mathbb{R}^3$ with zero first spatial derivatives is constant gives

$$P(\cdot, \tau) = a(\tau) \quad \text{in } \mathcal{D}'(\mathbb{R}^3)$$

for almost every τ . The function a is measurable because it may be chosen, for a fixed $\chi \in C_c^\infty(\mathbb{R}^3)$ with $\int \chi = 1$, as

$$a(\tau) = \langle P(\cdot, \tau), \chi \rangle.$$

This is an admissible pressure gauge. After subtracting it, $P - a(\tau) = 0$ almost everywhere, while $|V|^3 = 0$. Hence every nonvanishing integral of the form (2.2) or (2.3) is zero for the zero velocity field. $\square$

Lemma 2.11 (Basic properties). *Every Seregin ancient limit V satisfies*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty$$

and is smooth on $\mathbb{R}^3 \times \mathbb{R}$. After fixing the whole-space Leray pressure gauge, V is a bounded mild ancient solution of (1.3). Its time translates

$$V^s(y, \tau) = V(y, \tau + s), \quad s \in \mathbb{R},$$

are again bounded mild ancient solutions of (1.3) with the same L^∞ norm.

Proof. The L^∞ bound is the Type I ancient bound written in centered variables:

$$|V(y, \tau)| = \sqrt{-t} |U(y\sqrt{-t}, t)| \leq \sup_{s < 0} \sqrt{-s} \|U(s)\|_{L^\infty}.$$

Smoothness follows from the inherited Type I bound and standard local parabolic regularity for suitable weak solutions. Applying the Leray projection to (1.3) gives the whole-space projected Duhamel formulation

$$V(\tau) = S(\tau - \sigma)V(\sigma) - \int_{\sigma}^{\tau} S(\tau - s) \mathbb{P} \nabla \cdot (V \otimes V)(s) ds,$$

where S is the centered Stokes semigroup. This formulation gives a bounded mild representative. Since the centered equation is autonomous, $V^s(y, \tau) = V(y, \tau + s)$ is again a bounded mild ancient solution with the same L^∞ norm and introduces no additional normalization. $\square$

Lemma 2.12 (Local energy bounds in centered variables). *Let V be a Seregin ancient limit. For every $R < \infty$ and every compact interval $I \subset \mathbb{R}$, one has*

$$\int_I \int_{B_R} |V(y, \tau)|^2 dy d\tau + \int_I \int_{B_R} |\nabla V(y, \tau)|^2 dy d\tau < \infty.$$

If P is a pressure associated with V , then after subtracting a function of τ one also has

$$P \in L^{3/2}(B_R \times I).$$

These local bounds are stable under locally smooth limits of time translates on each fixed compact cylinder.

Proof. The unrenormalized ancient limit (U, Π) is suitable and satisfies the local energy and pressure bounds on compact negative-time cylinders by the extraction estimates in [Theorem 2.4](#). On every compact (y, τ) -cylinder the map

$$(y, \tau) \mapsto (x, t) = (e^{-\tau/2}y, -e^{-\tau})$$

has Jacobian $dx dt = e^{-5\tau/2} dy d\tau$, with $V = e^{-\tau/2}U$, $\nabla_y V = e^{-\tau}\nabla_x U$, and $P = e^{-\tau}\Pi$. Since the exponential weights are bounded above and below on compact τ -intervals, the local L^2 , dissipation, and $L^{3/2}$ -pressure estimates pass in the stated form. Stability under locally smooth velocity limits and weak local pressure limits is the following consequence of the same compactness package. If $V_n \rightarrow V$ in C_{loc}^∞ and the associated pressures, after admissible gauges, converge weakly in $L_{\text{loc}}^{3/2}$, then the L^2 and L^2 -gradient bounds pass on each $B_R \times I$ by ordinary convergence of the integrands $|V_n|^2$ and $|\nabla V_n|^2$, while the pressure bound passes by the weak lower semicontinuity of the $L^{3/2}(B_R \times I)$ -norm. $\square$

Remark 2.13. [Theorem 2.12](#) is deliberately local. It does not give uniform L^3 -tightness in y , nor does it give a global $L_\tau^\infty L_y^3$ bound. The absence of this information is the reason for the residual class below.

3. ADMISSIBLE CLASSES AND THE RESIDUAL CLASS

Let $\mathcal{S}$ denote a normalized collection of Seregin ancient limits in the sense of [Theorem 2.6](#). Thus all elements of $\mathcal{S}$ are associated with the same fixed nonvanishing normalization (K, η_0) , while their pressure gauges may differ. Every class introduced in this section is a subset of this fixed ambient set $\mathcal{S}$, and every set-theoretic complement is taken relative to $\mathcal{S}$. For the remainder of the paper, fix admissible parameter sets

$$I, J \subset (0, \infty),$$

where admissible means nonempty.

If $V \in \mathcal{S}$ is represented by an unrenormalized ancient solution U , then changing the extraction only within the same Seregin limit changes neither the velocity V nor the physical velocity U determined by

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

Thus the structural properties below are properties of the velocity field, not of the pressure gauge or of a particular pressure representative. All pointwise-in-time conditions in the definitions below are imposed on the smooth representative supplied by [Theorem 2.11](#). Hence statements

such as $V(y, \tau) = W(y)$, $L^\infty_\tau L^3_y$ -bounds, and uniform tail limits are literal statements for the smooth ancient solution, not merely assertions about a choice of time slices modulo a null set.

Definition 3.1 (Small-amplitude class). Let $\varepsilon_0 > 0$ be the absolute constant in the small bounded ancient Liouville theorem for (1.3). An element $V \in \mathcal{S}$ belongs to $\mathcal{X}_{\text{small}}$ if

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_0.$$

Definition 3.2 (Stationary L^3 class). An element $V \in \mathcal{S}$ belongs to $\mathcal{X}_{\text{stat-}L^3}$ if it is stationary in centered variables,

$$V(y, \tau) = W(y),$$

and

$$W \in L^3(\mathbb{R}^3).$$

Definition 3.3 (Uniformly tight class). An element $V \in \mathcal{S}$ belongs to $\mathcal{X}_{L^3\text{-tight}}$ if

$$(3.1) \quad \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty$$

and

$$(3.2) \quad \lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Definition 3.4 (Fast-decay class). Fix an admissible set $J \subset (0, \infty)$. An element $V \in \mathcal{S}$ belongs to $\mathcal{X}_{\text{fast}}(J)$ if, for some $m \in J$,

$$(3.3) \quad \sup_{\tau \in \mathbb{R}} \int_{\mathbb{R}^3} (1 + |y|^2)^m |V(y, \tau)|^2 dy < \infty.$$

Definition 3.5 (Structured class). Fix admissible parameter sets $I, J \subset (0, \infty)$. An element $V \in \mathcal{S}$ belongs to $\mathcal{X}_{\text{str}}(I, J)$ if its physical ancient representative U satisfies at least one of the following:

- (1) U is axisymmetric without swirl;
- (2) U is axisymmetric and, for some $\gamma \in I$,

$$\sup_{\tau \in \mathbb{R}} \sup_{\rho > 0, Z \in \mathbb{R}} \rho^\gamma |V_\theta(\rho, Z, \tau)| < \infty;$$

- (3) U is axisymmetric and, for some $1 \leq q < \infty$,

$$rU_\theta \in L_t^\infty L_x^q(\mathbb{R}^3 \times (-\infty, 0));$$

- (4) U is axisymmetric, periodic in the axial variable z , and

$$rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0));$$

- (5) V satisfies the weighted decay condition (3.3) for some $m \in J$.

Remark 3.6. The structured alternatives are the centered-limit formulation of the axisymmetric and weighted-decay branches. The stated properties are preserved by the self-similar transformation and by time translation because the axis of symmetry is unchanged, the swirl component scales by the same homogeneous factors as the velocity, axial periods scale in the physical variables and become stationary constraints in centered variables, and the weighted L^2 condition is imposed directly on $V(\tau)$. The no-swirl, $L_t^\infty L_x^p$ -swirl, periodic bounded- Γ , controlled-swirl, and weighted-decay exclusions are used below only through the single structured Liouville input stated below.

Definition 3.7 (Residual Seregin class). The residual class associated with $\mathcal{S}$, I , and J is

$$\mathcal{R}(\mathcal{S}; I, J) = \mathcal{S} \setminus (\mathcal{X}_{\text{small}} \cup \mathcal{X}_{\text{stat-}L^3} \cup \mathcal{X}_{L^3\text{-tight}} \cup \mathcal{X}_{\text{fast}}(J) \cup \mathcal{X}_{\text{str}}(I, J)).$$

Equivalently, $V \in \mathcal{R}(\mathcal{S}; I, J)$ is a Seregin ancient limit which is neither small in L^∞ , nor stationary and globally L^3 , nor uniformly L^3 -tight, nor fast decaying in an admissible weighted class, nor contained in the stated structured class.

Remark 3.8. The classes are not intended to be disjoint. For example, a weighted decay condition may imply an L^3 -tightness statement under additional boundedness and interpolation. The residual class is defined as the complement of the union, so overlaps among the non-residual classes cause no ambiguity.

Proposition 3.9 (Exhaustion by admissible classes). *Let $\mathcal{S}$ be a normalized collection of Seregin ancient limits, and fix admissible sets $I, J \subset (0, \infty)$. Every $V \in \mathcal{S}$ satisfies at least one of the following alternatives:*

$$\begin{aligned} V \in \mathcal{X}_{\text{small}}, & \quad V \in \mathcal{X}_{\text{stat-}L^3}, & \quad V \in \mathcal{X}_{L^3\text{-tight}}, \\ V \in \mathcal{X}_{\text{fast}}(J), & \quad V \in \mathcal{X}_{\text{str}}(I, J), & \quad V \in \mathcal{R}(\mathcal{S}; I, J). \end{aligned}$$

Proof. Set

$$\mathcal{U} = \mathcal{X}_{\text{small}} \cup \mathcal{X}_{\text{stat-}L^3} \cup \mathcal{X}_{L^3\text{-tight}} \cup \mathcal{X}_{\text{fast}}(J) \cup \mathcal{X}_{\text{str}}(I, J).$$

For any $V \in \mathcal{S}$, either $V \in \mathcal{U}$, in which case V belongs to at least one of the five displayed classes by the definition of union, or $V \notin \mathcal{U}$. In the second case

$$V \in \mathcal{S} \setminus \mathcal{U} = \mathcal{R}(\mathcal{S}; I, J)$$

by [Theorem 3.7](#). This proves the stated exhaustive alternative. $\square$

4. LIOUVILLE INPUTS FOR THE NON-RESIDUAL CLASSES

We now state the Liouville inputs used in the reduction. The first two closures are unconditional: small bounded ancient solutions vanish by the centered mild formulation, and stationary L^3 self-similar profiles vanish by the theorem of Necas–Ruzicka–Sverak. The remaining inputs are hypotheses of the present theorem, formulated so that each can be replaced by any independently proved Liouville theorem in the corresponding class.

Lemma 4.1 (Small bounded ancient Liouville theorem). *There is an absolute $\varepsilon_0 > 0$ such that every bounded mild ancient solution V of (1.3) with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_0$$

is identically zero.

Proof. Write the projected centered equation as

$$\partial_\tau V = LV - \mathbb{P} \nabla \cdot (V \otimes V), \quad L = \Delta - \frac{1}{2} y \cdot \nabla - \frac{1}{2}.$$

Let $S(s) = e^{sL}$ be the centered Stokes semigroup on divergence-free fields. The Ornstein–Uhlenbeck formula for $S(s)$ is

$$(S(s)f)(y) = e^{-s/2} \left(e^{(1-e^{-s})\Delta} f \right) (e^{-s/2} y), \quad s > 0.$$

Hence

$$\|S(s)f\|_{L^\infty} \leq e^{-s/2} \|f\|_{L^\infty}.$$

The corresponding projected-divergence estimate is the centered form of Kato's L^p estimate [5]. Thus there is an absolute constant C_0 such that, for every bounded tensor field F ,

$$\|S(s)\mathbb{P}\nabla \cdot F\|_{L^\infty} \leq C_0 e^{-s/2} (1 - e^{-s})^{-1/2} \|F\|_{L^\infty}, \quad s > 0.$$

The factor on the right is integrable on $(0, \infty)$. Set

$$A = C_0 \int_0^\infty e^{-s/2} (1 - e^{-s})^{-1/2} ds < \infty.$$

Indeed, $1 - e^{-s} \sim s$ as $s \downarrow 0$, so the kernel is comparable to $s^{-1/2}$ near the origin, while the factor $e^{-s/2}$ gives exponential decay as $s \rightarrow \infty$.

Fix $\tau \in \mathbb{R}$ and $T > 0$. The mild identity for the projected centered equation on $[\tau - T, \tau]$ gives

$$V(\tau) = S(T)V(\tau - T) - \int_0^T S(T - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - T + \sigma) d\sigma.$$

If $M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$, the preceding two estimates and the pointwise bound $\|V(\tau) \otimes V(\tau)\|_{L^\infty} \leq M^2$ imply

$$\begin{aligned} \|V(\tau)\|_{L^\infty} &\leq \|S(T)V(\tau - T)\|_{L^\infty} \\ &\quad + \int_0^T \|S(T - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - T + \sigma)\|_{L^\infty} d\sigma \\ &\leq e^{-T/2} M + C_0 M^2 \int_0^T e^{-(T-\sigma)/2} (1 - e^{-(T-\sigma)})^{-1/2} d\sigma. \end{aligned}$$

The change of variables $s = T - \sigma$ and the definition of A give

$$\|V(\tau)\|_{L^\infty} \leq e^{-T/2} M + AM^2.$$

Letting $T \rightarrow \infty$ yields $\|V(\tau)\|_{L^\infty} \leq AM^2$. Taking the supremum in τ gives $M \leq AM^2$. If $M > 0$, division by M gives $1 \leq AM$. Choose $\varepsilon_0 < 1/A$. The assumption $M \leq \varepsilon_0$ then contradicts $1 \leq AM$. Hence $M = 0$, which is equivalent to $V \equiv 0$. $\square$

Theorem 4.2 (Unconditional small and stationary branch exclusion). *Let $\mathcal{S}$ be a normalized collection of Seregin ancient limits, and let $V \in \mathcal{S}$ be nonzero. Then*

$$V \notin \mathcal{X}_{\text{small}} \quad \text{and} \quad V \notin \mathcal{X}_{\text{stat-}L^3}.$$

Proof. Assume first that $V \in \mathcal{X}_{\text{small}}$. By definition,

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_0,$$

where ε_0 is the small-data constant for bounded mild ancient solutions of the centered equation from Theorem 4.1. The mild solution property is part of Theorem 2.11, hence that lemma gives $V \equiv 0$. By Theorem 2.10, after subtracting the allowed pressure gauge the integral in (2.2) is zero. This contradicts the inherited positive normalization.

Assume next that $V \in \mathcal{X}_{\text{stat-}L^3}$. Then $V(y, \tau) = W(y)$ for some $W \in L^3(\mathbb{R}^3)$. Since V is a smooth distributional solution of (1.3), the identity $\partial_\tau V = 0$ is classical and the remaining terms in the equation are smooth functions of y . For every $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ and $\psi \in C_c^\infty(\mathbb{R})$, testing the equation against $\psi(\tau)\phi(y)$ gives

$$\int_{\mathbb{R}} \psi(\tau) \left\langle \nabla_y P(\cdot, \tau) - \Delta W + \frac{1}{2}(W + y \cdot \nabla W) + (W \cdot \nabla)W, \phi \right\rangle d\tau = 0.$$

Taking a countable dense family of spatial tests and using the fundamental lemma of distributions in τ , there is a full-measure set $E \subset \mathbb{R}$ such that, for every $\tau \in E$,

$$\nabla_y P(\cdot, \tau) = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W \quad \text{in } \mathcal{D}'(\mathbb{R}^3).$$

The right-hand side is independent of τ . Hence for $\tau_1, \tau_2 \in E$,

$$\nabla_y (P(\cdot, \tau_1) - P(\cdot, \tau_2)) = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^3),$$

so $P(\cdot, \tau_1) - P(\cdot, \tau_2)$ is spatially constant. Fix $\tau_0 \in E$, set $P_{\text{st}} = P(\cdot, \tau_0)$, and subtract the admissible time-dependent constant

$$a(\tau) = \langle P(\cdot, \tau) - P_{\text{st}}, \chi \rangle, \quad \int_{\mathbb{R}^3} \chi = 1,$$

for a fixed $\chi \in C_c^\infty(\mathbb{R}^3)$. Then the gauged pressure equals the single stationary pressure profile P_{st} for almost every $\tau \in E$, and the profile equation is

$$(W \cdot \nabla)W + \nabla P_{\text{st}} = \Delta W - \frac{1}{2}(W + y \cdot \nabla W), \quad \nabla \cdot W = 0.$$

The preceding calculation has identified the stationary mild trajectory with a stationary self-similar profile. The theorem of Nečas–Růžička–Šverák [9] then applies to the profile equation with $W \in L^3(\mathbb{R}^3)$, and gives $W \equiv 0$. Again [Theorem 2.10](#) makes the left-hand side of (2.2) vanish, contradicting the inherited normalization. $\square$

Hypothesis 4.3 (Tight-class Liouville input). For every normalized Seregin collection $\mathcal{S}$, no nonzero element of $\mathcal{S} \cap \mathcal{X}_{L^3\text{-tight}}$ exists.

Hypothesis 4.4 (Structured Liouville input). For every normalized Seregin collection $\mathcal{S}$, no nonzero element of $\mathcal{S} \cap \mathcal{X}_{\text{str}}(I, J)$ exists.

Remark 4.5. [Theorem 4.3](#) is the form obtained, for instance, from a minimal-element reduction plus compact-hull equilibrium rigidity in the uniformly L^3 -tight class. [Theorem 4.4](#) contains the structured reductions imported in [Theorem 3.6](#): the axisymmetric no-swirl Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [4], the $L_t^\infty L_x^p$ -swirl theorem of Lei–Zhang–Zhao [8], the periodic bounded- Γ theorem of Lei–Ren–Zhang [7], and any additional admissible controlled-swirl or weighted-decay Liouville criteria assumed for the sets I and J .

Proposition 4.6 (Non-residual class exclusion). *Assume [Theorems 4.3](#) and [4.4](#). Let $\mathcal{S}$ be a normalized collection of Seregin ancient limits, and let $V \in \mathcal{S}$ be nonzero. Then V cannot belong to*

$$\mathcal{X}_{\text{small}} \cup \mathcal{X}_{\text{stat-}L^3} \cup \mathcal{X}_{L^3\text{-tight}} \cup \mathcal{X}_{\text{fast}}(J) \cup \mathcal{X}_{\text{str}}(I, J).$$

Proof. The small-amplitude and stationary L^3 classes are excluded by [Theorem 4.2](#). Membership in $\mathcal{X}_{L^3\text{-tight}}$ is excluded by [Theorem 4.3](#). Membership in $\mathcal{X}_{\text{str}}(I, J)$ is excluded by [Theorem 4.4](#). Finally, $\mathcal{X}_{\text{fast}}(J) \subset \mathcal{X}_{\text{str}}(I, J)$ because both definitions require the existence of some $m \in J$ for which (3.3) holds, and this is precisely the fifth alternative in [Theorem 3.5](#). Thus the structured hypothesis also excludes the fast-decay class. $\square$

5. THE RESIDUAL CLASS

The residual class is the remaining alternative not covered by the stated Liouville inputs. We state the needed assertion in two equivalent forms.

Hypothesis 5.1 (No residual Seregin class). For the fixed admissible sets I, J , every finite-energy Type I source and every associated normalized collection $\mathcal{S}$ in the sense of [Theorem 2.7](#) satisfy

$$\mathcal{R}(\mathcal{S}; I, J) = \emptyset.$$

Hypothesis 5.2 (Residual rigidity). For the fixed admissible sets I, J , every finite-energy Type I source, every associated normalized collection $\mathcal{S}$ in the sense of [Theorem 2.7](#), and every element of $\mathcal{R}(\mathcal{S}; I, J)$ is identically zero.

Proposition 5.3 (Equivalence for Seregin limits). *For normalized Seregin collections associated with finite-energy Type I sources in the sense of [Theorem 2.7](#), the no-residual assertion [Theorem 5.1](#) and the residual rigidity assertion [Theorem 5.2](#) are equivalent.*

Proof. If [Theorem 5.1](#) holds, then the residual class is empty and the rigidity assertion is vacuous. Conversely, assume [Theorem 5.2](#), and let $\mathcal{S}$ be a normalized Seregin collection associated with a finite-energy Type I source in the sense of [Theorem 2.7](#). If $\mathcal{R}(\mathcal{S}; I, J) = \emptyset$, there is nothing to prove. If it is nonempty, choose $V \in \mathcal{R}(\mathcal{S}; I, J)$. Then [Theorem 5.2](#) gives $V \equiv 0$. By [Theorem 2.10](#), the associated pressure is spatially constant after gauge normalization, so the left-hand side of (2.3) is zero. This contradicts the defining lower bound $\eta_0 > 0$ of the normalized collection. Hence $\mathcal{R}(\mathcal{S}; I, J) = \emptyset$, which is [Theorem 5.1](#). $\square$

Remark 5.4. Analytically, [Theorem 5.1](#) may be proved in more than one way. One route is to show that finite energy and suitability force every Seregin limit to be uniformly L^3 -tight or fast decaying in self-similar variables. Another is to prove a Liouville theorem which applies directly to bounded ancient limits without tightness. Both routes are represented by the same formal statement above because either one eliminates $\mathcal{R}(\mathcal{S}; I, J)$.

6. CONDITIONAL TYPE I EXCLUSION

Theorem 6.1 (Residual-class criterion for Type I exclusion). *For the fixed admissible sets I, J , assume the tight-class Liouville input [Theorem 4.3](#), the structured Liouville input [Theorem 4.4](#), and the no-residual assertion [Theorem 5.1](#). Then no finite-energy suitable weak solution of the three-dimensional Navier–Stokes equations develops a Type I singularity in the sense of [Theorem 2.1](#).*

Proof. Suppose, toward a contradiction, that (u, p) has a finite-energy Type I singular point $z_* = (x_*, T)$. By [Theorem 2.4](#), there exists a Seregin ancient limit V associated with z_* , satisfying the inherited nonvanishing condition (2.2). This condition implies $V \not\equiv 0$: if $V \equiv 0$, then [Theorem 2.10](#) would make the left-hand side of (2.2) vanish. Fix the concentration normalization used to obtain this limit, and let $\mathcal{S}$ be the collection of all Seregin limits obtained from the same source with that fixed normalization. By [Theorems 2.6 and 2.7](#) and [Theorem 2.8](#), this $\mathcal{S}$ is an associated normalized collection; in particular it is nonempty because it contains the limit V .

By [Theorem 3.9](#), the following exhaustive assertion holds: either V belongs to at least one of the five non-residual classes, or

$$V \in \mathcal{R}(\mathcal{S}; I, J).$$

If V belongs to a non-residual class, then [Theorem 4.6](#) excludes it. This proposition invokes the unconditional small and stationary exclusions for the first two classes, [Theorem 4.3](#) for the uniformly L^3 -tight class, and [Theorem 4.4](#) for the structured and fast-decay classes. If V belongs to the residual class, then [Theorem 5.1](#) asserts that no such element exists. Both alternatives are impossible. This contradiction proves that the assumed Type I singular point cannot exist. $\square$

Corollary 6.2 (Replacement-rigidity version). *For the fixed admissible sets I, J , the conclusion of Theorem 6.1 remains valid if Theorem 5.1 is replaced by Theorem 5.2.*

Proof. Under Theorem 5.2, Theorem 5.3 gives Theorem 5.1 for the same fixed admissible sets I, J and the same associated normalized Seregin collections. The hypotheses of Theorem 6.1 are therefore satisfied, and the theorem gives the claimed exclusion of Type I singularities. $\square$

7. CONCLUSION

The criterion isolates the obstruction to finite-energy Type I exclusion as a single residual statement. The blow-up construction produces a nonzero Seregin ancient limit. The exhaustion result places this limit either in a class eliminated by the unconditional and assumed Liouville inputs or in the residual class. Thus the remaining analytic problem is to prove that finite-energy Seregin limits cannot persist in the residual non-tight/non-decaying regime, or to prove a Liouville theorem which applies to that regime directly.

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